

```
Atlas atlas-mzod4p-shard-0 [primary] bookflow_db> db.book_metadata.insertOne({
  title: "The Silent Algorithm",
  author: "R. Menon",
  genre: "Technology",
  publishedYear: 2022,
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    dir: true,
    narrator: "M. Rao",
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: "M1001",
      rating: 5,
      comment: "Excellent deep dive into digital systems design.",
      date: ISODate("2023-01-15")
    },
    {
      member_id: "M1042",
      rating: 4,
      comment: "Good but the Digital edition had formatting issues.",
      date: ISODate("2023-02-20")
    }
  ]
})

acknowledged: true,
insertedId: ObjectId('6a83d4a312db054b0bc24473')
```

```
Atlas atlas-mzod4p-shard-0 [primary] bookflow_db> db.book_metadata.insertMany([
  {
    title: "The Silent Algorithm",
    author: "R. Menon",
    genre: "Technology",
    publishedYear: 2022,
    formats: ["Hardcover", "Digital", "Audiobook"],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      dir: true,
      narrator: "M. Rao",
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: "M1001",
        rating: 5,
        comment: "Excellent deep dive into digital systems design.",
        date: ISODate("2023-01-15")
      },
      {
        member_id: "M1042",
        rating: 4,
        comment: "Good but the Digital edition had formatting issues.",
        date: ISODate("2023-02-20")
      }
    ]
  },
  {
    title: "The Silent Algorithm",
    author: "R. Menon",
    genre: "Technology",
    publishedYear: 2022,
    formats: ["Hardcover", "Digital", "Audiobook"],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      dir: true,
      narrator: "M. Rao",
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: "M1001",
        rating: 5,
        comment: "Excellent deep dive into digital systems design.",
        date: ISODate("2023-01-15")
      },
      {
        member_id: "M1042",
        rating: 4,
        comment: "Good but the Digital edition had formatting issues.",
        date: ISODate("2023-02-20")
      }
    ]
  },
  {
    title: "The Silent Algorithm",
    author: "R. Menon",
    genre: "Technology",
    publishedYear: 2022,
    formats: ["Hardcover", "Digital", "Audiobook"],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      dir: true,
      narrator: "M. Rao",
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: "M1001",
        rating: 5,
        comment: "Excellent deep dive into digital systems design.",
        date: ISODate("2023-01-15")
      },
      {
        member_id: "M1042",
        rating: 4,
        comment: "Good but the Digital edition had formatting issues.",
        date: ISODate("2023-02-20")
      }
    ]
  }
])

acknowledged: true,
insertedIds: [
  ObjectId('6a83d4a312db054b0bc24473'),
  ObjectId('6a83d4a312db054b0bc24474'),
  ObjectId('6a83d4a312db054b0bc24475')
]
```

```
Atlas atlas-mzod4p-shard-0 [primary] bookflow_db> db.book_metadata.find( { avgRating: 4.6 },
{
  _id: ObjectId('6a83d4a312db054b0bc24473'),
  title: "The Silent Algorithm",
  author: "R. Menon",
  genre: "Technology",
  publishedYear: 2022,
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    dir: true,
    narrator: "M. Rao",
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: "M1001",
      rating: 5,
      comment: "Excellent deep dive into digital systems design.",
      date: ISODate("2023-01-15")
    },
    {
      member_id: "M1042",
      rating: 4,
      comment: "Good but the Digital edition had formatting issues.",
      date: ISODate("2023-02-20")
    }
  ]
},
{
  _id: ObjectId('6a83d4a312db054b0bc24474'),
  title: "The Silent Algorithm",
  author: "R. Menon",
  genre: "Technology",
  publishedYear: 2022,
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    dir: true,
    narrator: "M. Rao",
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: "M1001",
      rating: 5,
      comment: "Excellent deep dive into digital systems design.",
      date: ISODate("2023-01-15")
    },
    {
      member_id: "M1042",
      rating: 4,
      comment: "Good but the Digital edition had formatting issues.",
      date: ISODate("2023-02-20")
    }
  ]
},
{
  _id: ObjectId('6a83d4a312db054b0bc24475'),
  title: "The Silent Algorithm",
  author: "R. Menon",
  genre: "Technology",
  publishedYear: 2022,
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    dir: true,
    narrator: "M. Rao",
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: "M1001",
      rating: 5,
      comment: "Excellent deep dive into digital systems design.",
      date: ISODate("2023-01-15")
    },
    {
      member_id: "M1042",
      rating: 4,
      comment: "Good but the Digital edition had formatting issues.",
      date: ISODate("2023-02-20")
    }
  ]
}
])
```

```
Atlas atlas-mzod4p-shard-0 [primary] bookflow_db> db.book_metadata.find( { formats: ["Digital", "Audiobook"] });
[
  {
    _id: ObjectId('64b5a0c3b0b0b0b0b0b0b0b0'),
    title: 'Ocean of Code',
    author: 'A. Kavanagh',
    genre: 'Science Fiction',
    publicationYear: 2020,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pageLen: 300,
      fileSize_MB: 0.2,
      min_year: 2010,
      max_year: 2020,
      runtime_min: 400
    },
    avgRating: 4.2,
    reviews: [
      {
        reviewer_id: 'R001',
        rating: 5,
        comment: 'A brilliant deep dive into digital system design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        reviewer_id: 'R002',
        rating: 5,
        comment: 'Good but the digital section had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('64b5a0c3b0b0b0b0b0b0b0b1'),
    title: 'The Silent Algorithm',
    author: 'B. Kavanagh',
    genre: 'Mystery',
    publicationYear: 2021,
    formats: [ 'Digital', 'Audiobook' ],
    specs: {
      pageLen: 350,
      fileSize_MB: 0.1,
      min_year: 2015,
      max_year: 2021,
      runtime_min: 450
    },
    avgRating: 4.6,
    reviews: [
      {
        reviewer_id: 'R003',
        rating: 5,
        comment: 'The digital format made annotations so easy.',
        date: ISODate('2023-03-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('64b5a0c3b0b0b0b0b0b0b0b2'),
    title: 'Quantum Horizons',
    author: 'C. Kavanagh',
    genre: 'Non-Fiction',
    publicationYear: 2022,
    formats: [ 'Digital', 'Audiobook' ],
    specs: {
      pageLen: 400,
      fileSize_MB: 0.3,
      min_year: 2018,
      max_year: 2022,
      runtime_min: 500
    },
    avgRating: 4.9,
    reviews: [
      {
        reviewer_id: 'R004',
        rating: 5,
        comment: 'Highly recommended book this year.',
        date: ISODate('2023-04-01T00:00:00.000Z')
      }
    ]
  }
]
```

```
Atlas atlas-mzod4p-shard-0 [primary] bookflow_db> db.book_metadata.find( { formats: ["Digital", "Audiobook"], runtime_min: 400 });
[
  {
    _id: ObjectId('64b5a0c3b0b0b0b0b0b0b0b0'),
    title: 'Ocean of Code',
    author: 'A. Kavanagh',
    genre: 'Science Fiction',
    publicationYear: 2020,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pageLen: 300,
      fileSize_MB: 0.2,
      min_year: 2010,
      max_year: 2020,
      runtime_min: 400
    },
    avgRating: 4.2,
    reviews: [
      {
        reviewer_id: 'R001',
        rating: 5,
        comment: 'A brilliant deep dive into digital system design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        reviewer_id: 'R002',
        rating: 5,
        comment: 'Good but the digital section had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('64b5a0c3b0b0b0b0b0b0b0b1'),
    title: 'The Silent Algorithm',
    author: 'B. Kavanagh',
    genre: 'Mystery',
    publicationYear: 2021,
    formats: [ 'Digital', 'Audiobook' ],
    specs: {
      pageLen: 350,
      fileSize_MB: 0.1,
      min_year: 2015,
      max_year: 2021,
      runtime_min: 450
    },
    avgRating: 4.6,
    reviews: [
      {
        reviewer_id: 'R003',
        rating: 5,
        comment: 'The digital format made annotations so easy.',
        date: ISODate('2023-03-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('64b5a0c3b0b0b0b0b0b0b0b2'),
    title: 'Quantum Horizons',
    author: 'C. Kavanagh',
    genre: 'Non-Fiction',
    publicationYear: 2022,
    formats: [ 'Digital', 'Audiobook' ],
    specs: {
      pageLen: 400,
      fileSize_MB: 0.3,
      min_year: 2018,
      max_year: 2022,
      runtime_min: 500
    },
    avgRating: 4.9,
    reviews: [
      {
        reviewer_id: 'R004',
        rating: 5,
        comment: 'Highly recommended book this year.',
        date: ISODate('2023-04-01T00:00:00.000Z')
      }
    ]
  }
]
```

```
Atlas atlas-mzod4p-shard-0 [primary] bookflow_db> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
);
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Quantum Horizons', avgRating: 4.9 }
]
```

```
Atlas atlas-mzod4p-shard-0 [primary] bookflow_db> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] },
  { title: 1, formats: 1, _id: 0 }
);
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  { title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
  { title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
]
```

```

Atlas atlas-wrodp-shard-0 [primary] bookflow_db db.book_metadata.find(
  {formats: ['$regex: "Digital"', '$options: "i" ]},
  {title: 1, formats: 1}
);
{
  _id: ObjectId('64b8b0e120e85b8bc3e079'),
  title: 'The Silent Algorithm',
  author: 'A. Person',
  genres: ['Science'],
  publicationYear: 2021,
  formats: ['Hardcover', 'Digital', 'Audiobook'],
  reviews: [
    {
      member_id: 'M001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: ISODate('2023-01-15T08:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('64b8b0e120e85b8bc3e079'),
  title: 'Ocean of Code',
  author: 'A. Fernandez',
  genres: ['Science'],
  publicationYear: 2021,
  formats: ['Digital', 'Paperback'],
  reviews: [
    {
      member_id: 'M001',
      rating: 5,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('64b8b0e120e85b8bc3e079'),
  title: 'Quantum Horizons',
  author: 'B. Person',
  genres: ['Science'],
  publicationYear: 2021,
  formats: ['Digital', 'Audiobook'],
  reviews: [
    {
      member_id: 'M001',
      rating: 5,
      comment: 'The Digital format made innovations so easy.',
      date: ISODate('2023-08-01T08:00:00.000Z')
    }
  ]
}

```

```

Atlas atlas-wrodp-shard-0 [primary] bookflow_db db.book_metadata.find(
  {formats: ['$regex: "Digital"', '$options: "i" ]},
  {title: 1, formats: 1}
);
{
  _id: ObjectId('64b8b0e120e85b8bc3e079'),
  title: 'The Silent Algorithm',
  formats: ['Hardcover', 'Digital', 'Audiobook']
},
{
  _id: ObjectId('64b8b0e120e85b8bc3e079'),
  title: 'Ocean of Code',
  formats: ['Digital', 'Paperback']
},
{
  _id: ObjectId('64b8b0e120e85b8bc3e079'),
  title: 'Quantum Horizons',
  formats: ['Digital', 'Audiobook']
}
Atlas atlas-wrodp-shard-0 [primary] bookflow_db db.book_metadata.find(
  {'reviews.comment': {'$regex: "Digital"', '$options: "i" }},
  {'title: 1, "reviews.$": 1, "_id: 0}
);
{
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: ISODate('2023-01-15T08:00:00.000Z')
    }
  ]
},
{
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M001',
      rating: 5,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    }
  ]
},
{
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M001',
      rating: 5,
      comment: 'The Digital format made innovations so easy.',
      date: ISODate('2023-08-01T08:00:00.000Z')
    }
  ]
}

```

```

Atlas atlas-wrodp-shard-0 [primary] bookflow_db db.book_metadata.aggregate(
  { $match: { publishedYear: { $gt: 2020 } } },
  { $unwind: '$reviews' },
  {
    $group: {
      _id: '$title',
      avgReviewRating: { $avg: '$reviews.rating' },
      totalReviews: { $sum: 1 },
      author: { $first: '$author' }
    }
  },
  { $sort: { avgReviewRating: -1 } },
  {
    $project: {
      _id: 0,
      title: '$_id',
      author: 1,
      avgReviewRating: { $round: [ '$avgReviewRating', 2 ] },
      totalReviews: 1
    }
  }
);
{
  totalReviews: 2,
  author: 'A. Person',
  title: 'Quantum Horizons',
  avgReviewRating: 5
},
{
  totalReviews: 2,
  author: 'A. Fernandez',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
},
{
  totalReviews: 2,
  author: 'A. Fernandez',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}

```

```

Atlas atlas-mzodp-shard-0 [primary] bookflow_db> db.book_metadata.find( { $text: { $search: "bigint" } } );
{
  "_id": ObjectId("6a83d9ea12db854db8bc2a473"),
  "title": "The Silent Algorithm",
  "author": "A. Karger",
  "genre": "Computer Science",
  "publisher": "MIT",
  "year": 2002,
  "pages": 288,
  "rating": 5,
  "date": "2002-01-01",
  "url": "http://www.mhhe.com/computer-science/books/9780262082000/0262082000.html",
  "isbn": "0262082000",
  "keywords": "Algorithms, Data Structures, Complexity Theory",
  "description": "A comprehensive introduction to the theory of algorithms, covering both classical and modern topics."
},
{
  "_id": ObjectId("6a83d9ea12db854db8bc2a474"),
  "title": "Ocean of Code",
  "author": "A. Karger",
  "genre": "Computer Science",
  "publisher": "MIT",
  "year": 2003,
  "pages": 300,
  "rating": 5,
  "date": "2003-01-01",
  "url": "http://www.mhhe.com/computer-science/books/9780262082000/0262082000.html",
  "isbn": "0262082000",
  "keywords": "Algorithms, Data Structures, Complexity Theory",
  "description": "A comprehensive introduction to the theory of algorithms, covering both classical and modern topics."
},
{
  "_id": ObjectId("6a83d9ea12db854db8bc2a475"),
  "title": "Quantum Horizons",
  "author": "A. Karger",
  "genre": "Computer Science",
  "publisher": "MIT",
  "year": 2004,
  "pages": 320,
  "rating": 5,
  "date": "2004-01-01",
  "url": "http://www.mhhe.com/computer-science/books/9780262082000/0262082000.html",
  "isbn": "0262082000",
  "keywords": "Algorithms, Data Structures, Complexity Theory",
  "description": "A comprehensive introduction to the theory of algorithms, covering both classical and modern topics."
}

```

```

Atlas atlas-mzodp-shard-0 [primary] bookflow_db> db.book_metadata.find(
  { $text: { $search: "bigint" } },
  { title: 1, score: { $meta: "textScore" } }
).sort( { score: { $meta: "textScore" } } );
{
  "_id": ObjectId("6a83d9ea12db854db8bc2a473"),
  "title": "The Silent Algorithm",
  "score": 1.1833333333333333
},
{
  "_id": ObjectId("6a83d9ea12db854db8bc2a474"),
  "title": "Ocean of Code",
  "score": 0.6
},
{
  "_id": ObjectId("6a83d9ea12db854db8bc2a475"),
  "title": "Quantum Horizons",
  "score": 0.6
}

```

```

Atlas atlas-mzodp-shard-0 [primary] bookflow_db> db.book_metadata.find(
  { $text: { $search: "bigint" } },
  { title: 1, score: { $meta: "textScore" } }
).sort( { score: { $meta: "textScore" } } );
{
  "_id": ObjectId("6a83d9ea12db854db8bc2a473"),
  "title": "The Silent Algorithm",
  "score": 1.1833333333333333
},
{
  "_id": ObjectId("6a83d9ea12db854db8bc2a474"),
  "title": "Ocean of Code",
  "score": 0.6
},
{
  "_id": ObjectId("6a83d9ea12db854db8bc2a475"),
  "title": "Quantum Horizons",
  "score": 0.6
}

```